

Bivariate Normal

STAT 24510-30040

(Winter 2025)

Data: n pairs $(x_1, y_1), \dots, (x_n, y_n)$, with $x_i, y_i \in \mathbb{R}, i = 1, \dots, n$.

Model assumptions: The pairs are independent observation of a pair of random variables (X, Y) .

For paired data, the interest is often on the relationship between the two variables. There are two main approaches.

- (1) We may look into the joint distribution of the pair, treating the two variables X and Y on equal footing.
- (2) We may view one of the variables Y as the response variable, then investigate how Y varies with the explanatory variable X by modeling the expected value of Y at each given value $X = x_i$.

This is the conditioning approach, which leads to linear regression models.

Of course the two approaches of paired data are closely related.

In this section, we take the approach of (1). We will investigate the most frequently used joint distribution for a pair of random variables, the bivariate normal distribution, and derive parameter estimations for bivariate normal data.

In addition, we bring in notations and properties of multivariate normal, which is essential to derive sample distribution and estimation theorems.

Bivariate Normal Properties

Review univariate normal random variables

Standard normal random variable $Z \sim (0, 1)$ has probability density function

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad z \in (-\infty, \infty) = \mathbb{R}$$

and cumulative distribution function

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(t) dt, \quad z \in (-\infty, \infty) = \mathbb{R}$$

Two parameters, the mean and the variance, completely determine the distribution of a normal random variable.

To construct a normal random variable X with mean a and variance b^2 from $Z \sim N(0, 1)$, we may let

$$X = a + bZ \quad \implies \quad X \sim N(a, b^2)$$

If $X \sim N(\mu, \sigma^2)$, then X can be written as a linear transformation of a standard normal, and X has density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$$

Review bivariate random variables

Consider bivariate random vector (X, Y) with

$$\mathbb{E}(X) = \mu_1, \quad \mathbb{E}(Y) = \mu_2, \quad \text{Var}(X) = \sigma_1^2, \quad \text{Var}(Y) = \sigma_2^2$$

In addition, the two random variables may be correlated,

$$\text{corr}(X, Y) = \rho$$

Recall the covariance and correlation of X and Y are

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mu_1)(Y - \mu_2)], \quad \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_1 \sigma_2}$$

If X and Y are continuous random variables, their joint probability density function $f(x, y)$ should incorporate all of the quantities and relationship stated above. Recall the two-variable cumulative distribution function is

$$F(x_o, y_o) = P(X \leq x_o, Y \leq y_o)$$

For continuous variables, their joint cumulative distribution and density function are related by

$$F(x_o, y_o) = \int_{-\infty}^{y_o} \int_{-\infty}^{x_o} f(x, y) dx dy, \quad \frac{\partial^2}{\partial x \partial y} F(x, y) = f(x, y)$$

Marginal density functions for each variable are obtained by integrating out the other variable from the joint density,

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

The conditional density functions are defined as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, \quad f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

with standard domain constraints such as the denominator is non-zero.

If we denote independence between X and Y as $X \perp\!\!\!\perp Y$, then

$$X \perp\!\!\!\perp Y \quad \iff \quad f(x, y) = f_X(x) f_Y(y), \quad \forall x, y \in \mathbb{R}$$

Bivariate normal density function

(X, Y) is **jointly normal** if their joint density is

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x-\mu_1}{\sigma_1} \right) \left(\frac{y-\mu_2}{\sigma_2} \right) + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right]}$$

for $(x, y) \in \mathbb{R}^2$, where parameters $\mu_1, \mu_2 \in \mathbb{R}$, $\sigma_1, \sigma_2 \in (0, \infty)$, and $\rho \in [-1, 1]$.

Basics of bivariate normal

- Bivariate normal is determined by five parameters

$$\mu_1, \quad \mu_2, \quad \sigma_1^2, \quad \sigma_2^2, \quad \rho$$

- Level curves of bivariate normal density on the (x, y) plane are ellipses with (μ_1, μ_2) as the center (exercise).
- To derive the marginal densities of each component, integrating out one variable in the joint density by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

The marginal distributions are obtained (exercises):

$$X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2)$$

Use the covariance definition and formula (exercises)

$$\text{cov}(X, Y) = \mathbb{E}(X - \mu_1)(Y - \mu_2) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

we can verify that

$$\text{cov}(X, Y) = \sigma_1 \sigma_2 \rho, \quad \text{corr}(X, Y) = \rho$$

Standardized bivariate normal

Similar to the univariate case where the standard normal $N(0, 1)$ makes it convenient to study the properties of general normal distributions, here we examine the standard form of bivariate normal.

Via variable transformation

$$U = \frac{X - \mu_1}{\sigma_1}, \quad V = \frac{Y - \mu_2}{\sigma_2}$$

we obtain the standardized bivariate normal vector (U, V) , with joint density

$$f(u, v) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(u^2 - 2\rho uv + v^2)}$$

The mean and variance parameters for the standard form are simplified to

$$E(U) = 0, \quad E(V) = 0, \quad \text{var}(U) = 1, \quad \text{var}(V) = 1$$

thus the marginal distributions are

$$U \sim N(0, 1), \quad V \sim N(0, 1)$$

while the scale-invariant correlation coefficient stays the same,

$$\text{cov}(U, V) = \text{corr}(U, V) = \rho.$$

Normal marginal distributions - a necessary condition

By integrating joint density function with respect to one variable, we can derive marginal density functions $f_U(u)$ and $f_V(v)$, $f_X(x)$ and $f_Y(y)$, respectively.

Each component random variable of a bivariate normal pair must have normal (marginal) distribution.

The converse is not true. That is, if both marginal distributions are normal, the joint distribution of the two normal random variables is not necessarily of bivariate normal.

If all linear combinations of the component variables are of normal distribution, then the joint distribution is a bivariate normal vector.

Conditional distributions of standard bivariate normal

First, consider conditional distributions for the standardized bivariate normal.

By definition, the conditional density of V given $U = u$ is

$$\begin{aligned} f(v|u) &= \frac{f(u, v)}{f_U(u)} = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}[u^2 - 2\rho uv + v^2]} \bigg/ \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \\ &= \frac{1}{\sqrt{2\pi(1-\rho^2)}} e^{-\frac{(v-\rho u)^2}{2(1-\rho^2)}} \end{aligned}$$

after algebraic simplifications. Note that, for fixed u , this function in v is the density function of $N(\rho u, 1 - \rho^2)$.

Conditional density of U given $V = v$ can be derived analogously.

The conditional distributions can be stated as

$$U|V=v \sim N(\rho v, 1 - \rho^2), \quad V|U=u \sim N(\rho u, 1 - \rho^2)$$

Consequently, the conditional expectations are

$$\mathbb{E}(U|V=v) = \rho v, \quad \mathbb{E}(V|U=u) = \rho u.$$

The conditional variances are

$$\text{var}(U|V=v) = 1 - \rho^2, \quad \text{var}(V|U=u) = 1 - \rho^2.$$

Remarks

Geometrically, for the joint density function of standardized bivariate normal on the (u, v) plane,

- The major axis of the elliptic level curves of the joint density of (U, V) is the line $u = v$ for $\rho > 0$.
- The center of the conditional distribution of V given $U = u$ is on the line $v = \rho u$.
- The center of the conditional distribution of U given $V = v$ is on the line $u = \rho v$, or $v = \frac{1}{\rho}u$.

Conditional distributions of general bivariate normal

For a general bivariate normal vector (X, Y) with

$$\mathbb{E}(X) = \mu_1, \quad \mathbb{E}(Y) = \mu_2, \quad \text{var}(X) = \sigma_1^2, \quad \text{var}(Y) = \sigma_2^2, \quad \text{corr}(X, Y) = \rho$$

(X, Y) can be related to standardized bivariate normal (U, V) through the transformation

$$\begin{cases} X = \sigma_1 U + \mu_1 \\ Y = \sigma_2 V + \mu_2 \end{cases}$$

By the conditional distribution results for (U, V) ,

$$\text{For given } u = \frac{x - \mu_1}{\sigma_1}, \quad \frac{y - \mu_2}{\sigma_2} = V \sim N(\rho u, 1 - \rho^2) = N\left(\rho \frac{x - \mu_1}{\sigma_1}, 1 - \rho^2\right)$$

Hence the conditional distribution of Y given $X = x$ is

$$Y|X = x \sim N\left(\frac{\sigma_2}{\sigma_1}\rho(x - \mu_1) + \mu_2, \sigma_2^2(1 - \rho^2)\right)$$

Analogously, the conditional distribution of X given $Y = y$ is

$$X|Y = y \sim N\left(\frac{\sigma_1}{\sigma_2}\rho(y - \mu_2) + \mu_1, \sigma_1^2(1 - \rho^2)\right)$$

Conditional expectations for general bivariate normal are

$$\mathbb{E}(X|Y = y) = \frac{\sigma_1}{\sigma_2}\rho(y - \mu_2) + \mu_1, \quad \mathbb{E}(Y|X = x) = \frac{\sigma_2}{\sigma_1}\rho(x - \mu_1) + \mu_2.$$

The conditional variances for general bivariate normal are

$$\text{var}(X|Y = y) = \sigma_1^2(1 - \rho^2), \quad \text{var}(Y|X = x) = \sigma_2^2(1 - \rho^2).$$

Remarks

Geometrically, on the (x, y) plane,

- The level curves of the joint density of (X, Y) are ellipses centered at (μ_1, μ_2) .
- The center of the conditional distribution of Y given $X = x$ are on the line passing (μ_1, μ_2) of slope $\frac{\sigma_2}{\sigma_1}\rho$.
- The center of the conditional distribution of X given $Y = y$ are on the line passing (μ_1, μ_2) of slope $\frac{\sigma_2}{\sigma_1}\frac{1}{\rho}$.
- The slope of the major axis of the level curve ellipses is complicated. If $\sigma_2 > \sigma_1$ and $\rho > 0$,

$$\tan \theta = \gamma + \sqrt{\gamma^2 + 1}, \quad \gamma = \frac{\beta - 1/\beta}{2\rho}, \quad \beta = \frac{\sigma_2}{\sigma_1}$$

- The slope = $\frac{\sigma_2}{\sigma_1}$ when $\rho = 1$ (not interesting) or when $\sigma_1 = \sigma_2$ and $\rho > 0$.

Random vectors and covariance matrices

Using vector-matrix notations, a vector

$$Z = \begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix} = [Z_1 \quad \cdots \quad Z_n]^T$$

is a random vector if the components $Z_1, \dots, Z_n$ are random variables with a multivariate joint distribution.

The mean vector and the variance-covariance matrix of a random vector is defined as

$$\mathbb{E}(Z) = \begin{bmatrix} \mathbb{E}(Z_1) \\ \mathbb{E}(Z_2) \\ \vdots \\ \mathbb{E}(Z_n) \end{bmatrix}, \quad \text{Cov}(Z) = \begin{bmatrix} \text{var}(Z_1) & \text{cov}(Z_1, Z_2) & \cdots & \text{cov}(Z_1, Z_n) \\ \text{cov}(Z_2, Z_1) & \text{var}(Z_2) & \cdots & \text{cov}(Z_2, Z_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(Z_n, Z_1) & \text{cov}(Z_n, Z_2) & \cdots & \text{var}(Z_n) \end{bmatrix}$$

Then, if $A = [a_{ij}]$ is a fixed matrix with scalar entries, it follows from the definitions of matrix operation that

$$\mathbb{E}(AZ) = A\mathbb{E}(Z), \quad \text{Cov}(AZ) = A\text{Cov}(Z)A^T$$

A^T denotes the transpose of matrix A .

Vector-matrix notation for normal vectors

The distribution of a joint normal vector can be expressed in terms of the mean vector and covariance matrix.

For bivariate normal (X, Y) we may use the vector-matrix notation

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim N_2\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right)$$

Denote

$$W = \begin{bmatrix} X \\ Y \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

Then we can simply write

$$W \sim N_2(\boldsymbol{\mu}, \Sigma)$$

The vector-matrix representation of the standard bivariate normal ($\mu_i = 0, \sigma_i^2 = 1$) has the simpler form

$$(U, V) \sim N_2\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right)$$

Σ is the variance-covariance or simply the covariance matrix of the random vector.

$$\Sigma = \text{Cov}(W) = \text{Cov}\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \text{var}(X) & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \text{var}(Y) \end{bmatrix}$$

Note that the inverse matrix

$$\Sigma^{-1} = \frac{1}{|\Sigma|} \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$$

where

$$|\Sigma| = \det(\Sigma) = \sigma_1^2\sigma_2^2(1 - \rho^2)$$

is the determinant of the matrix Σ .

Assuming non-degenerate cases with $|\Sigma| \neq 0$, then it can be verified that the joint density $f_W(w) = f_{X,Y}(x, y)$ can be expressed as

$$\frac{1}{2\pi|\Sigma|^{1/2}} e^{-\frac{1}{2}(w - \boldsymbol{\mu})^T \Sigma^{-1}(w - \boldsymbol{\mu})}$$

This expression can be generalized to multivariate normal density function with dimension > 2 .

Bivariate Normal Parameter Estimation

To derive parameter estimation methods, we assume the observed data are n independent pairs of random variables $(X_1, Y_1), \dots, (X_n, Y_n)$. For example, (X_i, Y_i) are the scores of a typical student i in two exams. Furthermore, assuming each pair is of joint normal,

$$(X_i, Y_i) \sim N_2 \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \right)$$

That is, for all $i = 1, \dots, n$,

$$\mathbb{E}(X_i) = \mu_1, \quad \mathbb{E}(Y_i) = \mu_2, \quad \text{Var}(X_i) = \sigma_1^2, \quad \text{Var}(Y_i) = \sigma_2^2, \quad \text{corr}(X_i, Y_i) = \rho.$$

Maximum Likelihood Estimation for i.i.d. joint normal pairs

The likelihood function given the data of n pairs $(X_i, Y_i), i = 1, \dots, n$ is a function of five parameters.

$$\ell = \ell(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho) = \ell\left(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho \mid (X_i, Y_i), i = 1, \dots, n\right)$$

By the independence between pairs,

$$\ell = f(x_1, y_1, x_2, y_2, \dots, x_n, y_n) = \prod_{i=1}^n f(x_i, y_i)$$

Thus

$$\ell = \left(\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \right)^n e^{-\frac{1}{2(1-\rho^2)} \sum_{i=1}^n \left[\left(\frac{x_i - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_i - \mu_1}{\sigma_1} \right) \left(\frac{y_i - \mu_2}{\sigma_2} \right) + \left(\frac{y_i - \mu_2}{\sigma_2} \right)^2 \right]}$$

The objective is to obtain a set of maximum likelihood estimates (MLE) for $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho$ which maximize the likelihood function.

The MLEs can be derived by differentiating $\log \ell$ with respect to the parameters $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho$, setting the partial derivatives to 0, then solving the five equations simultaneously (a tedious calculus exercise). In the derivation, it may help to complete the squares in the exponent, obtain the MLEs in the order of μ_i 's, σ_i 's, then ρ , plug in the derived estimators, cancel out sum-to-zero terms, and simplify during the process.

The derivation using the vector-matrix notation is more concise, thanks to the power of matrix algebra.

The ML estimators

The MLEs for the mean and variance vector are consistent with their corresponding one-dimensional, single sample MLEs. Sample means are the MLEs for the population means,

$$\hat{\mu}_1 = \bar{X}, \quad \hat{\mu}_2 = \bar{Y}$$

The variance MLEs are slightly different from the MSE sample variance S^2 ,

$$\hat{\sigma}_1^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S_1^2, \quad \hat{\sigma}_2^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{n-1}{n} S_2^2.$$

The MLE for ρ is the Pearson's product-moment correlation coefficient (also commonly denoted as R),

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = R \quad (1)$$

This is a familiar formula, since Pearson's product-moment correlation coefficient is widely used as the sample correlation coefficient formula.

A couple of equivalent expressions of the Person's correlation coefficient are

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})/n}{\sqrt{\hat{\sigma}_1^2 \hat{\sigma}_2^2}}$$

or

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})/(n-1)}{\sqrt{S_1^2 S_2^2}}$$

Density of $\hat{\rho} = R$ for bivariate normal

The statistic $R = \hat{\rho}$ is a function of $2n$ random variables $X_i, Y_i, i = 1, \dots, n$.

The derivation of the density function of R falls in the realm of multivariate theory. (Proof omitted; reference see chapter 4 in Anderson.)

When (X_i, Y_i) are i.i.d. observations from a bivariate normal vector, the probability density function of the correlation coefficient estimator $\hat{\rho}$ can be derived.

Even in the normal case, the density of $\hat{\rho} = R$ has a complex form involving infinite series or infinite order of derivatives.

$$f_R(r) = C_n (1 - \rho^2)^{\frac{n-1}{2}} (1 - r^2)^{\frac{n-4}{2}} \sum_{k=0}^{\infty} \frac{(2\rho r)^k \Gamma^2\left[\frac{1}{2}(n-1+k)\right]}{k!}$$

where the normalizing constant C_n can be expressed as

$$C_n = \frac{2^{n-3}}{\pi(n-1)!} = \frac{1}{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{n-1}{2}\right) \Gamma\left(\frac{n-2}{2}\right)}$$

When $\rho = 0$, the density of R simplifies to

$$f_R(0) = C_n (1 - r^2)^{\frac{n-4}{2}}$$

$\hat{\rho} = R$ is a slightly biased yet consistent estimator,

$$\mathbb{E}(R) = \rho - \frac{\rho(1-\rho^2)}{2n} + \dots$$

An approximation of the variance is

$$\text{Var}(R) \approx \frac{(1-\rho^2)^2}{n} \quad (2)$$

Properties of sample distribution of $\hat{\rho} = R$

- $\hat{\rho} \in [-1, 1]$.
- The density function $f_R(r)$ is symmetric when $\rho = 0$.
- The mode of the density function $f_R(r)$ skews to the right when $\rho > 0$, to the left when $\rho < 0$.
- The larger $|\rho|$, the more skewed its density function is.

Remarks

- Pearson's product-moment correlation coefficient R in (1) is commonly used as the default definition of sample correlation coefficient, regardless of the probability distribution of the underlying population.
- Given data, Pearson's correlation coefficient is often used as the sample estimate of the true correlation.
- All distribution theory for R and most usages and interpretations of sample correlation coefficient are based on the pre-assumption of a bivariate normal distribution.

Fisher's Transformation

The distribution of Pearson's correlation coefficient $R = \hat{\rho}$ in the formula in (1) is complicated. By (2), the variance of R varies with ρ (and n). For a given sample size n (thus treating n as constant),

$$\text{Var}(R) \approx \frac{(1-\rho^2)^2}{n} \propto (1-\rho^2)^2 = v(\rho)$$

Recall in the variance stabilizing transformation (VST) method, a variance stabilizing transformation $g(\rho)$ aims to have the property

$$\text{Var}(g(\rho)) \approx \text{constant}$$

which can be achieved by choosing

$$g'(\rho) = \frac{1}{\sqrt{v(\rho)}} = \frac{1}{1-\rho^2}$$

based on the delta method and the asymptotic unbiasedness $\mathbb{E}(\rho) \approx \rho$ for large n .

Via integration, the variance stabilizing transformation for $R = \hat{\rho}$ can be derived, with

$$g(\rho) = \int \frac{1}{1-\rho^2} d\rho = \frac{1}{2} \log \frac{1+\rho}{1-\rho}$$

Note that g is the inverse of the hyperbolic tangent function.

$$\frac{1}{2} \log \frac{1+\rho}{1-\rho} = \tanh^{-1} \rho = \text{arctanh}(\rho)$$

We have obtained the **Fisher transformation** of Pearson's correlation coefficient $\hat{\rho} = R$ in (1),

$$T = g(R) = \frac{1}{2} \log \frac{1+R}{1-R}$$

As a function of MLE $\hat{\rho} = R$, T is asymptotically of normal distribution. For moderately large n (say, > 5),

$$\frac{1}{2} \log \frac{1+R}{1-R} \sim N\left(\frac{1}{2} \log \frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right) \quad (\text{approximately})$$

where $n-3$ (instead of n) is obtained by finer approximation of the infinite series (proof omitted).

Confidence interval for correlation coefficient ρ

With Fisher's transformation, approximate confidence intervals can be obtained for

$$g(\rho) = \frac{1}{2} \log_e \frac{1+\rho}{1-\rho}$$

using the asymptotic distribution

$$T \sim N\left(\frac{1}{2} \log \frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right)$$

with

$$T = \frac{1}{2} \log \frac{1+R}{1-R} = g(R)$$

and the known quantiles of standard normal $N(0, 1)$.

Then a confidence interval for ρ can be derived by the conversion

$$R = g^{-1}(T) = \tanh(T) = \frac{e^{2T} - 1}{e^{2T} + 1}$$

Example

The observed sample correlation coefficient is $\hat{\rho} = 0.6$ with $n = 30$.

To obtain a confidence interval for ρ , consider the Fisher transformation $\theta = g(\rho) = \frac{1}{2} \log \frac{1+\rho}{1-\rho}$.

Fisher transformation evaluated at the sample correlation $\hat{\rho}$ is $g(\hat{\rho}) = \frac{1}{2} \log \frac{1+0.6}{1-0.6} = 0.69$.

By the asymptotic result,

$$\frac{1}{2} \log \frac{1+\hat{\rho}}{1-\hat{\rho}} \sim N\left(\frac{1}{2} \log \frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right)$$

approximately.

A 95% asymptotic confidence interval for

$$\theta = g(\rho) = \frac{1}{2} \log \frac{1+\rho}{1-\rho}$$

is

$$\left(0.69 - \frac{1.96}{\sqrt{30-3}}, 0.69 + \frac{1.96}{\sqrt{30-3}}\right) = (0.31, 1.07)$$

To convert an confidence interval for $\rho = g^{-1}(\theta)$, use the monotone property of

$$g^{-1}(\theta) = \frac{e^{2\theta} - 1}{e^{2\theta} + 1}$$

At the endpoints of the confidence interval

$$g^{-1}(0.31) = 0.30, \quad g^{-1}(1.07) = 0.79$$

Thus we obtain

$$95\% \text{ asymptotic confidence interval for } \rho = (0.30, 0.79)$$

Relevant sections in Rice: 3.3 - 3.6, 4.2 - 4.4; Casella and Berger chapter 4.